

Digital Practices and Digital Memories

Bibliometric records and the representation of academic work

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TRANSUR — Work Package B

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Opening

Why we are in a room together

- ▶ **Groundwork:** laying out a set of principles and resources for communal use. Or, *commoning*, as a principle and a governance practice.
- ▶ The common(s) are **produced, not given**. They are social relations before they are things.
- ▶ For this week we aim to: build a base for collaborative work to thrive and find out, together, what type of empirical claims we can/want to work on to surface **South-to-North circulation**.

Dardot & Laval (2014); Caffentzis & Federici (2014). We will work on this throughout Friday.

The argument of this workshop

- ▶ Representations of the world through data result from **work, organizational routines, people's practices and specific technologies** that articulate them. Hence: infrastructures.
- ▶ Data is brought to life, made possible and real, through **specific material actions**.
- ▶ We look closely at one type of data: **bibliometric data**, which is itself **metadata created to support the circulation of texts and ideas**.
- ▶ We look for creative and critical ways to **work with data** (e.g., parsing, repurposing, reprocessing, etc.).

The argument, continued

- ▶ Networks are **at once** technological constructions, a set of specific associations, and a method to chart relations.
- ▶ Like classification systems and the memory practices they uphold, they are both the **result of** and the **means by which** specific power dynamics are sustained.
- ▶ They are also effective and functional arrangements that support circulation (and bear traces of it).
- ▶ So representation through (meta)data tends to favor **what works**: it makes legible what already has a role in reproducing the knowledge system (mirroring hegemony and centrality).

Venturini, Munk & Jacomy (2019).

What we are trying to do

- ▶ Digital tools let us **repurpose, reconfigure and reorganize** these representations.
- ▶ That is what allows **objects to speak**, and to express the relations that uphold them in sociotechnical terms.
- ▶ **Legibility** is not something data *has*. It is something you bring to the surface through specific operations and a given set of concepts.

The workshop's question

How can we use records, along with digital tools working in connection to in-depth qualitative accounts, to repurpose bibliographic metadata and unearth South-to-North circulations?

The week ahead

The WPB workflow¹: two phases

- ▶ **FASE 1 — building and validating search inputs**

WPB_MECILA WPB_KEYWORD-diccio WPB_AUTORES-diccio

- ▶ **FASE 2 — retrieving data with those validated inputs**

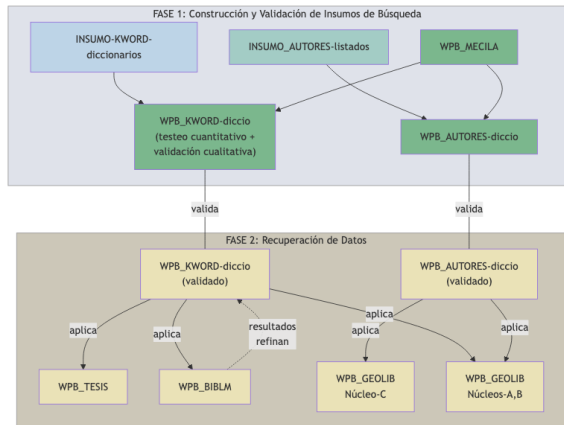
WPB_TESIS WPB_GEOLIB WPB_BIBLM

- ▶ WPB_BIBLM feeds newly discovered terms **back** into WPB_KEYWORD-diccio.
The pipeline has a loop, not just a direction.

0_doc_v0-9.ipynb, "Diagrama del Flujo de Trabajo".

¹We use workflow, but sometimes also the idea of "pipeline", as it differs from workflow in the sense that it is automated and results from a fixed set of instructions and processes.

The workflow, as documented



Module architecture, dependencies and feedback loops

What each module produces

- ▶ **WPB_MECILA** — metadata, content descriptors and author list from the Cuadernos de MECILA collection
- ▶ **WPB_KEYWORD-diccio** — the validated keyword dictionary
- ▶ **WPB_AUTORES-diccio** — validated author-name variants
- ▶ **WPB_TESIS** — thesis corpus, France and Germany
- ▶ **WPB_GEOLIB** — geolocated library holdings
- ▶ **WPB_BIBLM** — the OpenAlex bibliometric corpus this workshop works with

0_doc_v0-9.ipynb, "Tipos de Módulos".

Three traditions, one set of tools

- ▶ **Computational Social Sciences** — existing data as a proxy for social phenomena
- ▶ **Digital Humanities** — navigating archives, making sense of extensive digital records
- ▶ **Digital Methods** — objects branded by their digital way of existence

Tools circulate across all three. The differences are in what each tradition takes the digital *to be*.

Milia (2023b).

Tailors and thrift stores

Custom-made

Built from scratch for your question. Precise, and limited by money, time, staff, and by what is still reachable: you cannot survey a past event.

Ready-made

The product of pre-existing processes. Not made for your question. Might not be exactly what you want, but can be good enough. Requires creativity to re-process, interpret and visualize.

Bibliometric data is ready-made.

Everything we do this week follows from that.

Milia (2023b); Salganik, *Bit by Bit*, ch. 2.

A digital research workflow

1. Identify sources of suitable data
2. Query, merge or parse databases
3. Extract information from texts
4. Appropriate data to answer research questions
5. Determine results and ask new questions

Unlike an automated pipeline, it requires **human judgment at every step**.

Milia (2023b). This is the inspiration for how the week itself is structured.

An ideal workflow

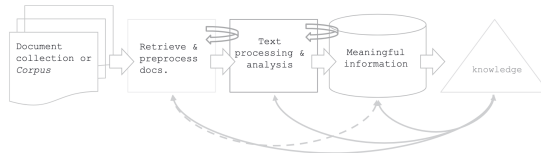


Figure 44.2 Example of an ideal workflow for digital text analysis

Milia (2023b)

Scientometrics: where we start, not where we stand

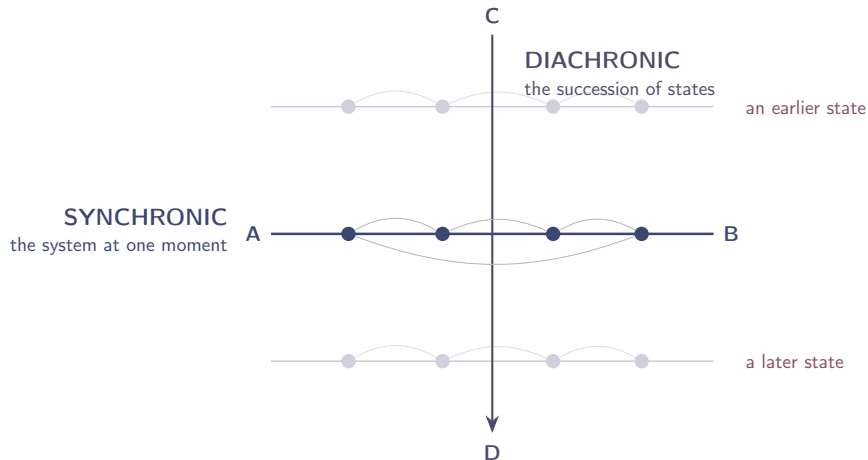
Two axes

- ▶ **Diachronic** — traces change and transformation across time. Accounts for long-lasting processes with historical depth.
- ▶ **Synchronic** — looks at structure and relationships within a system at a single point in time. Provides meaningful categories and descriptions.

We can analyze bibliographic information to articulate both or focus on each of them.

Operational definition following Saussure's early formulations. Workshop program.

How the two axes meet



Any analysis sits at the crossing: one state, at one moment, in a series.

A field with a shared origin

- ▶ Quantitative methods applied to the formal and semantic aspects of scientific literature, treated as information and communication processes.
- ▶ Scientometrics and qualitative STS **share a common origin** — and have grown apart in research practices, norms and standards.

Wyatt et al. (2017), p. 87.

The field was built alongside its data

- ▶ **1955** — Garfield proposes citation indexing in *Science*, modeled on Shepard's Citations, an index in the legal domain since 1873.
- ▶ **1955–60** — the citation index as an invention.
- ▶ **1961** — the Science Citation Index published experimentally, after the founding of the ISI.
- ▶ **late 1980s** — citation indexing reconsidered: no longer only a retrieval tool, now advocated as a **research instrument**.
- ▶ **1978** — the journal *Scientometrics* founded.

Leydesdorff (2010).

The map is performative

- ▶ The scientometric map is a real analytical achievement: it makes patterns visible that no amount of close reading would reach.
- ▶ It also **appears objective** — while the assumptions that produce it sit in parameter choices that are not always accessible to the reader.

Wyatt et al. (2017), p. 97.

Where we go from here

- ▶ Scientometrics has been **critical in the establishment of science studies**.
- ▶ Still, proper social research on knowledge circulation **cannot rest on scientometric techniques alone**.
- ▶ It requires **triangulation with ethnographic work on infrastructures**, complementing traditional ethnographic techniques in STS with methods from scientometrics and text analysis.
- ▶ It can also **benefit from creative interactions** with other research strategies (e.g., archival work, close-reading, interviews, etc.)

Venturini, Munk & Jacomy (2019), p. 511; Wyatt et al. (2017), p. 87.

Articulating with the rest of TRANSUR

- ▶ **Work Package C — Translation:** translation flows between Latin America, France and Germany. Presence/absence of works in languages, spread in translations, etc.
- ▶ **Work Package A — Institutionalization:** the institutionalization and transformation of Latin American studies in France and Germany. The infrastructures that support this.
- ▶ Which lets us ask: what is the role of institutions in supporting the technologies that make texts retrievable — or fail to?

TRANSUR project document, p. 9, p. 12. Task TB 4 foresees WP A and WP C insights discussed at the annual workshop.

One person's daily work

One person's daily work is another person's infrastructure.

Information infrastructure is a **relational** concept. Nothing is infrastructure in itself; it becomes infrastructure in relation to organized practices.

Bowker et al. (2010), p. 98, citing Star & Ruhleder (1996).

Doing digitality takes work

- ▶ Digital things are the **contingent result of a process of association** — one which is itself aiming to reduce the contingency of digital productions.
- ▶ That work is what turns traces into metrics and indicators, through specific ways of **parsing and connecting** data sources.

Infrastructures as **assemblages** (addition, design, intention) or as **intersections** (contingent, indeterminate).

Camus & Vinck (2019), p. 3.

Always already cooked

- ▶ Data are never “raw”. They are always already **cooked** by the disciplines that collect, store and process them.
- ▶ To talk about data is, at bottom, to talk about **how each discipline conceives its own objects**.

Gitelman & Jackson (2013), p. 3–4, p. 8, drawing on Bowker.

In the dataverse

- ▶ It makes no sense to speak of the raw and the cooked, the natural and the social.
- ▶ Both are flattened into a **single regime** of agents or actants sharing the same computable properties.

Bowker (2013), "Data Flakes", p. 4.

Memory practices and metadata

What bibliometric data is for

Bibliometric data is what allows retrieval.

It runs on **persistence** and **aggregation**. That makes it a crucial practice supporting memory and retrieval — and this is the sense in which our data is, nonetheless, **metadata created to support the circulation of texts and ideas**.

Four things metadata is

1. A **feature** of information systems
2. A **result of labor**
3. A critical part of **memory practices**
4. For us: the item that lets us **map emerging research trends by detecting features**

The drive to database

Databases are not a product of the computer revolution.

If anything, the computer revolution is a product of the drive to database.

Bowker (2008), p. 109.

Documenting competes with doing

- ▶ Empirical studies have shown, time and again, that people will not see it as a good use of their time to preserve information about their data beyond what guarantees its **immediate usefulness**.
- ▶ The case in point: the medical culture of quick, easy diagnoses on **death certificates** — written to meet the administrative need and free time to get on to the next (live) patient.

Bowker (2008), p. 10, citing Fagot-Largeault (1989).

Handshakes

- ▶ Federating databases — parsing disparate efforts, establishing **handshakes** between information systems — is what makes these memory practices powerful.
- ▶ Bowker's formulation: strings of bits traveling along wires are **meaningless** unless there is a shared set of handshakes.
- ▶ Standards setting is a **key site of political work**.

Bowker (2008), p. 113–114.

Memory got cheap

- ▶ Memory prostheses have multiplied. We now hold in global memory details of events hundreds of thousands of years ago, and masses of detail about lives in the twentieth century.
- ▶ Memory teases apart into two distinct acts: **remembering what happened** and **making present again**.
- ▶ Digital transformation multiplies both the volume retained **and** the challenge of reading and interpreting it.

Bowker (2008), p. 110.

The past lives on through classifications

- ▶ On Bowker and Star's account, access to the past runs through classification systems — formal or informal, hierarchical or not.
- ▶ Time makes past forms inaccessible except as mediated by **systematization and standardization**.
- ▶ “We are seeking to colonize the past just as we are seeking to colonize the present.”

Bowker & Star (1999), p. 9; Bowker (2008), p. 108.

Metadata, defined²

Metadata (“data about data”) is the technical term for all the information a single piece of data can carry with it, in order to provide sufficient context for another user to first **locate** it and then **use** it.

Standardized metadata — Dublin Core, FGDC — is what lets you search a given kind of data, say authors, across multiple databases.

Bowker (2008), p. 10.

²Metadata is Tuesday morning’s main subject: metadata as inscription — made by someone, for someone, to be read by someone else — data architectures, what each database documents and why, and annotation. Today we need only enough of it to follow this session’s argument.

Context costs work

- ▶ You always have to know **some** context.
- ▶ The more information you provide to make data useful to the **widest community** over the **longest time**, the more work you have to do.
- ▶ This is the first difficulty in developing an information infrastructure — and we just saw why people don't do it.

Bowker (2008), p. 10.

An aside: what metadata alone reveals

- ▶ From **telephone metadata alone** — no call content — researchers inferred participants' likely medical conditions and firearm ownership.
- ▶ Calls to a cardiologist, a drugstore and an arrhythmia monitoring hotline. That was enough.

Mayer, Mutchler & Mitchell (2016), *PNAS*. A decade old, and still the clearest demonstration.

Data work is never neutral

- ▶ What circulates as bibliometric data is always already constituted within wider **data assemblages** — technological, political, social, economic.
- ▶ The point to stress is the **value of the data work**, not the volume of the data.

Iliadis & Russo (2016), p. 1–2.

Classification is narrow, and efficient

- ▶ Classifications are a central part of the built worlds we inhabit, including the scientific.
- ▶ Scientometric analysis **leans on classifications**. These are narrow — but the organization of scientific work they produce is **efficient**.
- ▶ Much of the materiality of bibliometric data is deeply connected to that efficiency.
- ▶ What we see represented from a paper is an **artifact** that gains value and sense from classification itself.

Bowker & Star (1999), "Some Tricks of the Trade", p. 33–50.

The practical politics of classifying

- ▶ Whatever appears as universal, or indeed standard, is the **result of negotiations**, organizational processes and conflict.
- ▶ Large-scale classification systems are often **invisible**, erased by their naturalization into the routines of life.

Bowker & Star (1999), p. 45, p. 48.

The convergence principle

- ▶ Systems shape what can be known, because methods determine what can be observed.
- ▶ Classifications become **self-fulfilling**.

“This blindness occurs by changing the world such that the system’s description of reality becomes true.”

Bowker & Star (1999), p. 50.

Easier to use, harder to see

- ▶ The easier a classification system is to use, and the more widely it is adopted, the **harder it becomes to see**.
- ▶ Classifications become invisible as they sink into infrastructure.
- ▶ Concretely, for us: in bibliometric databases, and in the digital representations of scientific publications themselves.

Bowker & Star (1999), p. 1.

Three networks, one word

- ▶ A conceptual topology (the space of connections)
- ▶ A set of computational techniques (the mathematics of graphs)
- ▶ The hypertextual organization of inscriptions (relational datasets)

Digital inscriptions are **not created by or for the social sciences**. They are the product of vast sociotechnical systems, and bring the influence of those systems with them.

Venturini, Munk & Jacomy (2019), p. 2, p. 514.

What metadata brands

Metadata is never neutral

- ▶ Metadata is a **designed instrument**: it exists to make things findable, and it works.
- ▶ Which is why it always makes a **statement about the world** — subjective in what it includes, what it omits, where it draws its boundaries, and the terms it uses to describe.
- ▶ Non-neutrality here is a property of an instrument built for a purpose, not a defect in it.

Gartner (2016), p. 4.

A history of control

On Acker's reading, every history of metadata is, at bottom, a history of asserting **control** and designating **access** through representation.

Control and access are also what make a collection usable at all. The claim is about what metadata *does*, not about anyone's bad intentions.

Acker (2021), p. 1.

Knowledge infrastructures

How do we know?

An established fact is one supported by an infrastructure.

Without infrastructure there are no claims you can back up, no facts you can verify, and no comprehension you can share.

Edwards (2010), p. 20, p. 23.

Infrastructural inversion

“To understand an infrastructure, you have to invert it. You turn it upside down and look at the ‘bottom’ — the parts you don’t normally think about precisely because they have become standard, routine, transparent, invisible.”

Edwards (2010), p. 21.

Knowledge infrastructures

- ▶ Interconnected sets of **people, institutions and technologies**.
- ▶ Critical systems that let memories exist in specific, social and material terms.
- ▶ **Consensus holders**: common understandings of how we know what we know.
- ▶ “For us, infrastructures reside in a naturalized background... we notice them mainly when they fail.”

Edwards (2010), p. 9, p. 20.

Making global data / making data global

Making global data

Gathering records scattered across time and space into one usable corpus. The effort this takes has been called **data friction**.

Making data global

Building models that adjust for the unevenness of those observations, to produce a single coherent global data image.

“Virtually any global thing you try to study will bring you up against these issues.”

Edwards (2010), p. xiv–xv, p. xix.

Our case

- ▶ Article metadata in **OpenAlex** is aggregated from multiple sources.
- ▶ This pair of concepts is what lets us disentangle what we are actually seeing represented in the BIBLM corpus.
- ▶ Tomorrow we retrieve it and look.

On OpenAlex coverage: Céspedes et al. (2025) — more balanced linguistic coverage than Web of Science, but language metadata is not always accurate.

Documents as inscriptions

- ▶ Documents do not merely **represent** the world. They refer back to the practices, objects, rules, knowledge and organizational forms that **produced** them.
- ▶ Read as **inscriptions** of scientific work, in the Latourian sense.
- ▶ Both a **noun** (the artifact) and a **verb** (the process of documenting) — a tension that exists because documenting competes with the actual doing.
- ▶ Documents hold still enough to **travel and be reused**, and are read differently in each setting they reach. That **fixity and fluidity** is what makes them usable as evidence, and what makes them demanding to use.
- ▶ This is the payoff for social research: because a document carries the conditions of its own production, it lets us study practices we cannot observe directly. A bibliographic record is a **document about a document**.

Shankar et al. (2017), p. 4–6.

Making objects speak

- ▶ By performing actions on these data, digital tools open the possibility of **repurposing, reconfiguring and reorganizing** the representations themselves.
- ▶ That is what allows for **objects to speak**, and to express the relations that uphold them in sociotechnical terms.

Latour (2007), "How to Keep the Social Flat?", pp. 165–173; Callon (2006), *European Management Review*, 3(1), 7–16.

“We need to open a discourse — where there is no effective discourse now — about the varying temporalities, spatialities and materialities that we might represent in our databases, with a view to designing for maximum flexibility and allowing as much as possible for an emergent **polyphony** and **polychrony**. Raw data is both an oxymoron and a bad idea; to the contrary, data should be **cooked with care**.”

Bowker (2008), p. 182.

What “legible” means

Not something data *has*.

Something you bring to the surface through specific operations, and through a given set of concepts and theorizations.

Hermeneutic expertise, in critical, sociotechnical and infrastructural terms.

Technical and hermeneutic

- ▶ Critical data analysis combines **technical skill** with **interpretive capacity**, across the whole data cycle.
- ▶ These can go hand in hand — and they should, if what you want is a **practical, working critique** of existing accounts of how texts, and the ideas authors embed in them, circulate as part of a wider epistemic landscape.

Which returns us to the question

How can we use records, digital tools and in-depth qualitative accounts together, to repurpose bibliographic metadata and unearth South-to-North circulations?

Planted for Friday

The common, briefly

- ▶ The common is a **verb**, not a noun: the practice or process of **commoning**, not a property of things.
- ▶ **Co-obligation** made by participation: obligation exists between those who participate in the same activity, not on the basis of belonging or membership.
- ▶ Not the same as **the public**: the public is subtracted from appropriability by an act of law; the common attains that status through its nature.
- ▶ Only **practical activity** can make the common.

Dardot & Laval (2019). Recovered on Friday.

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